

Project 3 — SQL Data Analysis

DecodeLabs Data Analytics Industrial Training Program
Batch 2026

1. Project Overview

For this project I used SQL Server to dig into an e-commerce order dataset and see what it says about the business. Everything was written and run in SQL Server Management Studio (SSMS) against a single table, **dbo.DecodeLabs**, which holds 1,200 orders.

The analysis covers orders, products, customers, order status, time trends, payment methods, referral sources, coupons and basket behaviour. I began with data quality checks so I could trust the numbers, and then worked my way from the overall figures down to products, statuses, time, customers and order sizes.

The full SQL script is a separate file, *project3_analysis.sql*. This report sticks to the main queries, what they returned, and what I think the results mean for the business.

A quick note on terms. Throughout this report, **Order Value** means the sum of TotalPrice across all orders, whatever their status (Delivered, Shipped, Pending, Cancelled or Returned). Because cancelled and returned orders are included, it is not money the business actually collected. A few SQL outputs label the same measure “Revenue”, but in the write-up I always call it Order Value.

2. Project Objectives

- Check the quality of the dataset
- Understand overall business performance
- Compare how the products perform
- Look at cancelled and returned orders
- Study monthly and yearly trends
- Find high-value and repeat customers
- Compare payment methods and referral sources
- Analyse coupon usage
- Examine order-size and basket patterns

3. Dataset Description

The dataset is an e-commerce transaction table with 1,200 orders, 1,189 unique customers and 7 products: Chair, Desk, Laptop, Monitor, Phone, Printer and Tablet. Each row is one order. Order dates run from January 2023 to June 2025, so 2025 covers only six months.

Column	Meaning
Order ID	Unique ID of the order
Date	Date the order was placed
Customer ID	ID of the customer who placed the order
Product	Product ordered (7 different products)
Quantity	Number of units in the order
Unit Price	Price of one unit
Total Price	Total price of the order; used as the order value
Payment Method	How the order was paid (Credit Card, Online, Cash, Gift Card, Debit Card)
Order Status	Delivered, Shipped, Pending, Cancelled or Returned

Column	Meaning
Tracking Number	Shipment tracking number (not used in the analysis)
Items in Cart	Number of items in the cart for that order; used in the basket analysis
Coupon Code	Coupon used on the order: FREESHIP, SAVE10, WINTER15 or NO_COUPON
Referral Source	Where the customer came from (Instagram, Email, Google, Facebook, Referral)
Shipping Address	Delivery address (not used in the analysis)

Table 1. Columns in the dataset and what they mean.

4. Tools Used

SQL Server stored the dataset and ran every query, all written in T-SQL.

SQL Server Management Studio (SSMS) was where I wrote and ran the queries and looked at the result grids. I saved the output to a results file (*Project3_All_Query_Results.rpt*), and all the numbers in this report come from that file.

5. Data Quality Checks

Before analysing anything, I ran three checks on the table. All three came back clean.

Check	What the query looks for	Result
Duplicate Order IDs	Order IDs that appear more than once	0 rows returned
Missing values	NULLs in OrderID, CustomerID, Product, Date, Quantity, UnitPrice and TotalPrice	0 missing in all seven columns
Total price check	Orders where TotalPrice differs from Quantity × UnitPrice by more than 0.01	0 rows returned

Table 2. Data quality checks and their results.

The raw prices show long decimals in the output (for example 570.619995117188), so in the third check I allowed a difference of 0.01 to avoid flagging tiny rounding differences. The missing-value check only covered the seven columns listed above. Since nothing turned up, no cleaning was needed and I used all 1,200 rows.

```
-- Check for duplicate orders
SELECT
    OrderID,
    COUNT(*) AS DuplicateCount
FROM dbo.DecodeLabs
GROUP BY OrderID
HAVING COUNT(*) > 1;
```

Duplicate Order ID check (from project3_analysis.sql).

```

7  -----
8  -- 1. DATA QUALITY CHECKS
9  -----
10
11
12  -- Check for duplicate orders
13
14  SELECT
15      OrderID,
16      COUNT(*) AS DuplicateCount
17  FROM dbo.DecodeLabs
18  GROUP BY OrderID
19  HAVING COUNT(*) > 1;
20
21
22  -- Check missing values in important columns
23
24  SELECT
25      SUM(CASE WHEN OrderID IS NULL THEN 1 ELSE 0 END) AS MissingOrderID,
26      SUM(CASE WHEN CustomerID IS NULL THEN 1 ELSE 0 END) AS MissingCustomerID,
27      SUM(CASE WHEN Product IS NULL THEN 1 ELSE 0 END) AS MissingProduct,
28      SUM(CASE WHEN Date IS NULL THEN 1 ELSE 0 END) AS MissingDate,

```

110 % No issues found Ltr: 51, Chr: 1 SPC CRLF UTF-8

OrderID	DuplicateCount
1	0

MissingOrderID	MissingCustomerID	MissingProduct	MissingDate	MissingQuantity	MissingUnitPrice	MissingTotalPrice
0	0	0	0	0	0	0

OrderID	Quantity	UnitPrice	TotalPrice	ExpectedTotal
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Figure 1. SSMS results for the data quality checks: no duplicate OrderIDs, no missing values in any column, and no rows where TotalPrice disagrees with the expected total.

6. SQL Analysis

6.1 Overall Business Performance

I started with a sample query to get a feel for the table, then ran a summary query for the headline figures, and finally pulled the highest-value delivered orders.

Measure	Value
Total orders	1,200
Unique customers	1,189
Products	7
Units sold	3,535
Total Order Value	₹12,64,761.96
Average Order Value	₹1,053.97
Smallest order	₹11.39
Largest order	₹3,456.40

Table 3. Overall business summary.

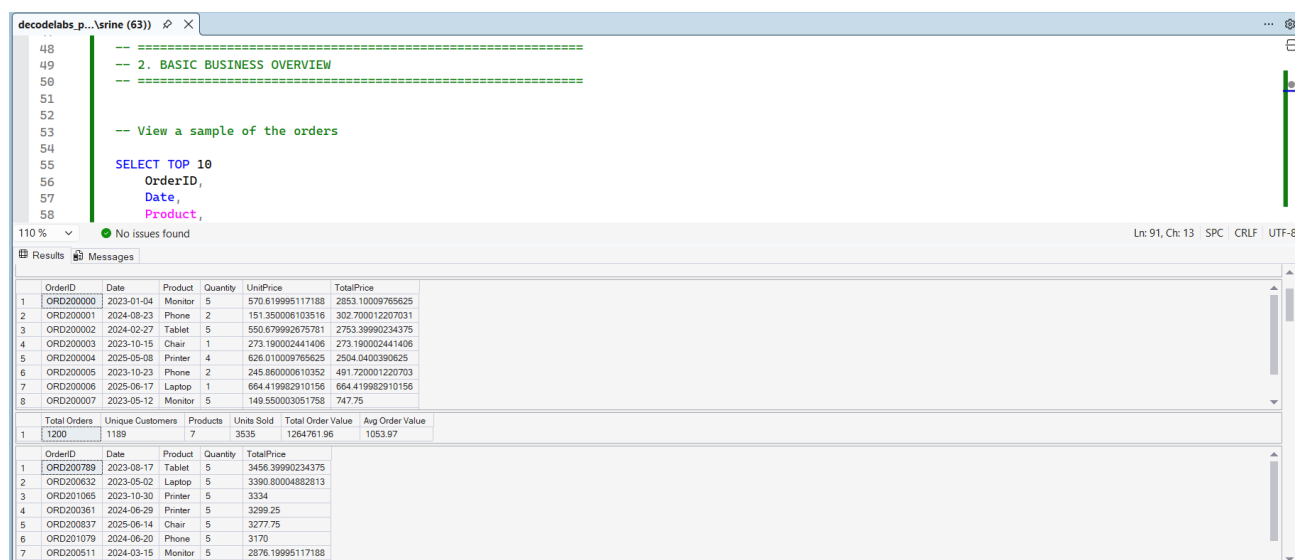


Figure 2. SSMS output for the sample of orders, the overall summary row, and the highest-value delivered orders.

The business received 1,200 orders from 1,189 customers, and the average order came to about ₹1,054. Order sizes vary a lot, from ₹11.39 up to ₹3,456.40. Among delivered orders, the biggest was ORD200789 (Tablet, 5 units, ₹3,456.40), which is also the largest order in the whole table. Nine of the top 10 delivered orders had a quantity of 5, and Printer showed up four times in that list. So the largest orders mostly come from people buying several units, not from one very expensive item.

6.2 Product Analysis

Here I grouped orders by product and compared orders, units, Order Value and average order value. I also worked out each product's share of the total and ranked the products with RANK().

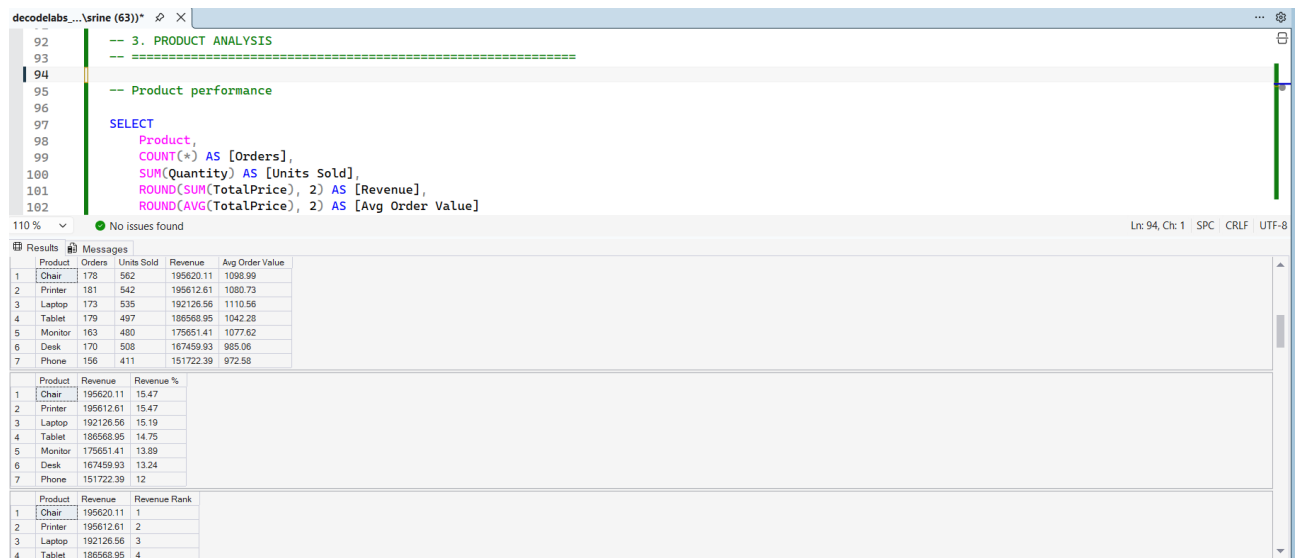


Figure 3. SSMS output for product performance, each product's share of total Order Value, and the RANK() result (the third grid is cut off after rank 4 in the screenshot).

Product	Orders	Units	Order Value	Avg Order Value	Share %	Rank
Chair	178	562	₹1,95,620.11	₹1,098.99	15.47	1
Printer	181	542	₹1,95,612.61	₹1,080.73	15.47	2
Laptop	173	535	₹1,92,126.56	₹1,110.56	15.19	3
Tablet	179	497	₹1,86,568.95	₹1,042.28	14.75	4

Product	Orders	Units	Order Value	Avg Order Value	Share %	Rank
Monitor	163	480	₹1,75,651.41	₹1,077.62	13.89	5
Desk	170	508	₹1,67,459.93	₹985.06	13.24	6
Phone	156	411	₹1,51,722.39	₹972.58	12.00	7

Table 4. Product performance, share of total Order Value and rank.

Chair came out on top with ₹1,95,620.11 (15.47%), but Printer is just ₹7.50 behind at ₹1,95,612.61, so the two are effectively tied. Phone was the weakest, with ₹1,51,722.39 (12.00%) and the fewest orders (156) and units (411). Laptop had the highest average order value (₹1,110.56) even though it sits third on total value. Overall the shares only range from 12% to 15.47%, so no single product carries the business.

6.3 Order Status Analysis

I looked at orders and Order Value by status, then used a CASE expression to work out the share of orders that were Cancelled or Returned. I repeated that calculation for each product, and finished with a query on the Order Value of Shipped and Delivered orders only.

Order Status	Orders	Order Value	Avg Order Value
Cancelled	250	₹2,76,396.21	₹1,105.58
Returned	247	₹2,43,277.70	₹984.93
Pending	237	₹2,56,328.15	₹1,081.55
Shipped	235	₹2,46,159.58	₹1,047.49
Delivered	231	₹2,42,600.32	₹1,050.22

Table 5. Orders and Order Value by status.

Of the 1,200 orders, 497 were Cancelled or Returned, which is 41.42%. The orders are spread almost evenly over the five statuses, and only 231 were Delivered, fewer than Cancelled (250) or Returned (247). Cancelled orders even carry the highest Order Value of any status (₹2,76,396.21), which tells me a lot of the total Order Value never became a completed sale.

Product	Orders	Cancelled or Returned	Loss Rate %	Shipped + Delivered Value
Monitor	163	71	43.56	₹66,924.44
Tablet	179	77	43.02	₹66,622.67
Laptop	173	74	42.77	₹80,951.66
Chair	178	73	41.01	₹71,816.27
Printer	181	73	40.33	₹82,452.14
Phone	156	62	39.74	₹62,342.05
Desk	170	67	39.41	₹57,650.67

Table 6. Cancellation/return rate by product, with the Order Value of Shipped and Delivered orders. "Loss Rate %" is the column name used in the SQL output.

Monitor had the highest cancellation/return rate (43.56%), followed by Tablet (43.02%). The gap between products is small, though. Even the lowest, Desk, sits at 39.41%, so this looks like a store-wide issue and not a problem with just two products. When I count only Shipped and Delivered orders, Printer comes first (₹82,452.14) and Laptop second (₹80,951.66), while Chair, the top product overall, slips to third (₹71,816.27).

6.4 Time Analysis

I used YEAR() and MONTH() to group orders by year and month, and a CTE with a window function to build a running (cumulative) Order Value by month.

Year	Orders	Order Value	Avg Order Value
2023	510	₹5,52,643.24	₹1,083.61
2024	459	₹4,80,235.87	₹1,046.27
2025 (Jan-Jun)	231	₹2,31,882.85	₹1,003.82

Table 7. Yearly performance.

Month	2023	2024	2025
Jan	56,685.75 (47)	38,528.08 (32)	29,099.40 (27)
Feb	40,117.66 (37)	36,909.57 (32)	35,317.55 (37)
Mar	48,609.37 (43)	36,030.90 (36)	39,200.66 (49)
Apr	27,751.71 (31)	49,613.14 (50)	31,821.20 (32)
May	63,836.84 (49)	27,909.11 (34)	43,396.64 (37)
Jun	49,500.19 (45)	68,068.54 (53)	53,047.40 (49)
Jul	42,820.66 (44)	42,963.98 (43)	-
Aug	54,352.14 (51)	31,991.07 (28)	-
Sep	29,526.67 (29)	39,794.98 (44)	-
Oct	52,607.85 (47)	37,226.97 (31)	-
Nov	43,079.67 (41)	32,413.76 (35)	-
Dec	43,754.73 (46)	38,785.77 (41)	-

Table 8. Monthly Order Value in ₹ (number of orders in brackets).

June 2024 was the strongest month, with ₹68,068.54 from 53 orders. The weakest was April 2023 at ₹27,751.71 from 31 orders. Monthly order counts stayed between 27 and 53 the whole time. The average order value fell every year, from ₹1,083.61 in 2023 to ₹1,046.27 in 2024 and ₹1,003.82 in 2025. Because 2025 only has six months, its total can't be compared directly with the full years 2023 and 2024. The cumulative query ends at ₹12,64,761.96 in June 2025 (₹5,52,643.24 by December 2023 and ₹10,32,879.11 by December 2024), which matches the overall total, so that was a nice sanity check.

6.5 Customer Analysis

I listed the top 10 customers by total spent, then the customers with more than one order (using HAVING), and finally worked out how much of the Order Value came from repeat customers (using a CTE).

Customer ID	Orders	Total Spent	Avg Order Value
C38840	2	₹5,723.23	₹2,861.61
C57276	1	₹3,456.40	₹3,456.40
C67260	1	₹3,390.80	₹3,390.80
C13877	1	₹3,384.90	₹3,384.90
C18404	1	₹3,370.20	₹3,370.20
C16775	1	₹3,353.75	₹3,353.75

Customer ID	Orders	Total Spent	Avg Order Value
C65986	1	₹3,352.40	₹3,352.40
C47778	1	₹3,334.00	₹3,334.00
C59183	1	₹3,322.55	₹3,322.55
C25276	1	₹3,313.90	₹3,313.90

Table 9. Top 10 customers by total spent.

Only 11 customers placed more than one order, and every one of them placed exactly two (which fits with 1,200 orders from 1,189 customers). Repeat customers made up just 1.54% of total Order Value. C38840 is the top spender with ₹5,723.23 from two orders, and every other customer in the top 10 got there with a single large order, between ₹3,313.90 and ₹3,456.40. In this data, then, a “high-value customer” mostly means someone who made one big purchase, and almost nobody came back to buy again.

```
WITH CustomerOrders AS
(
  SELECT CustomerID, COUNT(*) AS Orders, SUM(TotalPrice) AS TotalSpent
  FROM dbo.DecodeLabs
  GROUP BY CustomerID
)
SELECT
  ROUND(100.0 * SUM(CASE WHEN Orders > 1 THEN TotalSpent ELSE 0 END)
  / SUM(TotalSpent), 2) AS [Repeat Customer Revenue %]
FROM CustomerOrders;
```

Repeat customer share of Order Value (same logic as in project3_analysis.sql, shown on fewer lines).

6.6 Payment and Referral Analysis

Both queries group the orders (by payment method and by referral source) and compare orders, Order Value and average order value. For referral sources I also used a subquery to get each source’s share of total Order Value.

Payment Method	Orders	Order Value	Avg Order Value
Credit Card	234	₹2,63,847.63	₹1,127.55
Online	258	₹2,62,442.94	₹1,017.22
Cash	246	₹2,59,786.29	₹1,056.04
Gift Card	230	₹2,46,323.92	₹1,070.97
Debit Card	232	₹2,32,361.18	₹1,001.56

Table 10. Payment method performance.

Referral Source	Orders	Order Value	Avg Order Value	Share %
Instagram	259	₹2,75,285.45	₹1,062.88	21.77
Email	250	₹2,61,808.55	₹1,047.23	20.70
Google	241	₹2,50,441.48	₹1,039.18	19.80
Facebook	228	₹2,50,410.90	₹1,098.29	19.80
Referral	222	₹2,26,815.58	₹1,021.69	17.93

Table 11. Referral source performance and share of total Order Value.

Credit Card had the highest Order Value (₹2,63,847.63) and the highest average order (₹1,127.55). Online had the most orders (258) but a lower average order, and Debit Card was lowest on both Order Value and average order. Overall the differences between payment methods are fairly small. On the referral side, Instagram

brought in the most Order Value, ₹2,75,285.45 or 21.77% of the total, along with the most orders (259). Facebook had fewer orders (228) but the highest average order (₹1,098.29). Google and Facebook are almost level on total value, and the “Referral” source came last at 17.93%. One limitation: these queries didn’t look at order status, so they show how much value each source brought in but not how much of it was later cancelled or returned.

6.7 Coupon Analysis

I grouped the orders by coupon code, then used HAVING with a subquery to find the coupon groups whose average order value was above the overall average of ₹1,053.97.

Coupon Code	Orders	Order Value	Avg Order Value
FREESHIP	313	₹3,35,036.99	₹1,070.41
NO_COUPON	309	₹3,22,401.41	₹1,043.37
WINTER15	292	₹3,02,483.54	₹1,035.90
SAVE10	286	₹3,04,840.02	₹1,065.87

Table 12. Coupon performance.

FREESHIP was the most-used coupon with 313 orders, just ahead of orders with no coupon (309). Only two groups beat the overall average order value: FREESHIP (₹1,070.41) and SAVE10 (₹1,065.87). WINTER15 had the lowest average (₹1,035.90), even lower than orders with no coupon at all. So coupon orders aren’t clearly bigger than normal ones. The dataset has no discount amount, so I couldn’t check what the coupons actually cost.

6.8 Order Value Analysis

I used a CASE expression to sort orders into four groups: Low Value (below ₹500), Medium Value (₹500 to below ₹1,500), High Value (₹1,500 to below ₹2,500) and Very High Value (₹2,500 and above). These cut-offs are my own choice for this analysis.

Order Category	Orders	Order Value	Avg Order Value
Low Value	383	₹1,01,696.90	₹265.53
Medium Value	495	₹4,53,172.33	₹915.50
High Value	228	₹4,40,179.79	₹1,930.61
Very High Value	94	₹2,69,712.94	₹2,869.29

Table 13. Orders grouped by order value.

Medium Value orders formed the biggest group (495 orders, ₹4,53,172.33). The 383 Low Value orders make up almost a third of all orders but only about 8% of the Order Value. At the other end, the 94 Very High Value orders brought in ₹2,69,712.94, roughly a fifth of the total. A small number of big orders clearly matters a lot here.

6.9 Yearly Product Performance

To find the top product in each year, I used a CTE for yearly Order Value per product and then ROW_NUMBER() with PARTITION BY year, keeping only rank 1.

```
ROW_NUMBER() OVER (
  PARTITION BY [Year]
  ORDER BY Revenue DESC
) AS ProductRank
-- ... then filtered with WHERE ProductRank = 1
```

Part of the yearly product query (the full version is in project3_analysis.sql).

Year	Top Product	Order Value
2023	Printer	₹95,513.73
2024	Chair	₹84,785.75
2025 (Jan-Jun)	Desk	₹47,291.58

Table 14. Highest-value product in each year.

The top product changed every year: Printer in 2023, Chair in 2024 and Desk in 2025. Chair is first over the whole period but only led in 2024, and Desk is sixth overall yet led the first half of 2025. One caveat: ROW_NUMBER() returns exactly one product per year, so a tie wouldn't show up, although nothing in the results suggests one.

6.10 Basket Analysis

This part compares the quantity ordered with the number of items in the cart. I calculated the average of both by product and counted the orders where Quantity equals ItemsInCart.

Product	Orders	Avg Quantity	Avg Items in Cart
Chair	178	3.16	5.78
Desk	170	2.99	5.58
Monitor	163	2.94	5.56
Printer	181	2.99	5.51
Tablet	179	2.78	5.50
Laptop	173	3.09	5.34
Phone	156	2.63	5.08

Table 15. Average quantity and average items in cart by product.

Only 200 orders (16.67%) had Quantity equal to ItemsInCart. For every product the average items in cart (5.08 to 5.78) is higher than the average quantity ordered (2.63 to 3.16), so in most orders the quantity didn't match the cart size. Chair is highest on both measures and Phone is lowest on both. The dataset doesn't say what the other cart items were, so I'm treating this purely as an observation about basket size and not as proof of what customers actually did.

7. Key Findings

- 1,200 orders from 1,189 unique customers across 7 products.
- Total Order Value was ₹12,64,761.96, with an average order of ₹1,053.97.
- Chair had the highest overall Order Value at ₹1,95,620.11 (Printer was very close at ₹1,95,612.61).
- 41.42% of orders were Cancelled or Returned (497 of 1,200).
- Monitor had the highest cancellation/return rate at 43.56%, with Tablet at 43.02%.
- June 2024 was the highest-value month: ₹68,068.54 from 53 orders.
- Repeat customers contributed only 1.54% of total Order Value.
- Instagram generated ₹2,75,285.45, which is 21.77% of total Order Value among referral sources.
- FREESHIP was the most-used coupon, with 313 orders.
- The 94 Very High Value orders generated about ₹2,69,712.94.
- Top product by year: Printer in 2023, Chair in 2024, Desk in 2025.
- 200 orders (16.67%) had Quantity equal to ItemsInCart.

8. Business Recommendations

These suggestions come only from what the queries showed.

- **Find out why orders are cancelled or returned.** 497 of 1,200 orders (41.42%) fell into this group, but the dataset has no reason column, so the cause will have to come from other records.
- **Look closely at products with high cancellation/return rates.** Monitor (43.56%) and Tablet (43.02%) are the highest, but every product is above 39%, so general causes like delivery or order handling should be checked as well.
- **Work on retention and repeat purchases.** Only 11 customers ordered twice and repeat customers gave 1.54% of Order Value, so there is plenty of room to bring buyers back.
- **Judge referral sources on order value and order quality.** Instagram brings in the most Order Value, but the status of those orders should be checked too.
- **Use yearly product performance for inventory and promotion planning.** The top product changed every year (Printer, Chair, Desk), so recent data matters. Keep in mind that the 2025 figures only cover January to June.
- **Target high-value customers with suitable offers or cross-selling.** The 94 Very High Value orders produced ₹2,69,712.94, so keeping those buyers interested is worthwhile.
- **Evaluate coupon campaigns on more than usage.** FREESHIP is the most used, but its average order is only slightly above the overall average. Discount cost and order status by coupon should be compared as well.

9. SQL Concepts Used

Concept	Where it was used
SELECT	All queries
WHERE	Delivered orders, Shipped/Delivered orders, total price check, Quantity = ItemsInCart
ORDER BY	Sorting results by Order Value, orders, rate, year and month
GROUP BY	Product, status, year, month, customer, payment method, referral source, coupon
HAVING	Duplicate check, repeat customers, coupons above the overall average
COUNT()	Orders, unique customers (COUNT(DISTINCT)), duplicates
SUM()	Order Value, units sold, running totals
AVG()	Average order value, average quantity, average items in cart
CASE	Missing-value counts, cancelled/returned flag, order value categories
Subqueries	Share of total Order Value, coupon average vs overall average, percentage of baskets
CTEs	Cumulative monthly value, repeat customer share, yearly product ranking
Window Functions	Cumulative SUM() OVER, RANK(), ROW_NUMBER() with PARTITION BY
RANK()	Ranking products by Order Value
ROW_NUMBER()	Top product in each year
Date-based analysis	YEAR() and MONTH() for yearly and monthly trends
Data-quality validation	Duplicate, missing value and total price checks

Table 16. SQL concepts and where they appear in the analysis.

I also used TOP, ROUND, CAST and COUNT(DISTINCT) along the way.

10. Learning Outcome

This project gave me practice using SQL to answer real business questions instead of just running isolated syntax exercises. I got more comfortable with GROUP BY and CASE for summaries, subqueries for percentages, and CTEs and window functions for running totals and yearly rankings.

It also taught me that a single number can mislead. Chair has the highest Order Value, but once I counted only Shipped and Delivered orders it dropped to third. The data quality checks, and the care I took to call the measure Order Value and not revenue, came from the same lesson: be clear about what a number really includes. There is still a lot more for me to learn, but this was a good step toward using SQL on transaction data.

11. Conclusion

In this project I went beyond basic SQL syntax and used queries to explore real business questions on an e-commerce dataset of 1,200 orders. After the data passed all quality checks, the analysis showed a total Order Value of ₹12,64,761.96, a high share of cancelled and returned orders (41.42%), very few repeat customers (1.54% of Order Value), and a product mix where no single product dominates. I see these as useful starting points for further investigation, and the full queries are in *project3_analysis.sql*.

Appendix — Query Categories

Category	Type of queries	Report section
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Product Analysis	Product performance queries	6.2
Order Status	Cancellation/return analysis	6.3
Time Analysis	Monthly/yearly trends	6.4
Customer Analysis	Customer and repeat-order analysis	6.5
Payment/Referral	Source performance	6.6
Coupon	Coupon usage and AOV	6.7
Order Value	Value segmentation	6.8
Yearly Product	Window-function analysis	6.9
Basket Analysis	Quantity vs ItemsInCart	6.10

Table 17. Query categories in *project3_analysis.sql*.